

Common Knee Joint Problems

The Big Picture

The knee is a **hinge synovial joint** between the two longest bones — bony shape gives little stability, so **ligaments are the main stabilisers**. Topics: traumatic dislocations, extensor apparatus ruptures, **meniscal tears**, osteochondritis dissecans, loose bodies, deformities, and **osteoarthritis** (the knee is the commonest large joint for OA). Use the **diagnostic calendar** (age → likely diagnosis).

Assessment

- History: trauma, **pain, catching, clicking, locking**. Bloods: CBC/ESR, FBS, uric acid.
- Imaging: X-ray (**AP standing**, lateral, axial, intercondylar notch), CT (fractures), **MRI (ligaments)**, bone scan, arthrography.

Patellar Dislocation

- Normal **valgus** alignment → patella tends to dislocate **laterally** when quads contract.
- Mechanism: indirect (flexed knee + quads) or direct lateral blow.
- Tearing sensation, falls; patella usually relocates, sometimes stays as lateral lump.
- Treat: **conservative relocation 2–3 weeks**; surgery if medial patellofemoral ligament ruptured.

Knee Dislocation (emergency)

- High-energy (RTA/fall) → ruptures **ACL, PCL, LCL, MCL** ± fracture; haemarthrosis.
- **40% popliteal vessel injury, 20% common peroneal nerve injury** — limb-threatening.
- Treat: **immediate reduction** (avoid hyperextension — protects popliteal vessel), check circulation x48 h; vascular injury → urgent surgery + **ex-fix**; staged arthroscopic ligament reconstruction.

Extensor Apparatus Disruption (age-based)

- Adolescent → **avulsion of tibial tubercle**.
- Young adult → **patellar tendon rupture** (high-riding patella; prior steroid injection a risk).
- Middle-aged → **patella fracture**.
- Older/chronic illness → **quadriceps tendon rupture**.
- Test: if patient **can actively extend**, extensor mechanism intact.

Meniscal Lesions

- Menisci improve congruency/stability, control roll-glide, **distribute load**; removal ↑ articular stress.
- **Medial meniscus** less mobile → torn more often than lateral.
- **Tear** from rotational grinding (flexed + twisted under load) — footballers; minor force in older fibrosed meniscus.
- **Avascular except outer third** (only it can heal).
- Features: medial pain, **locking (bucket-handle tear)**, delayed swelling, giving way.
- Treat: acute → rest/splint/ice 3–4 wk; **arthroscopic meniscectomy or repair** if locked/recurrent. Asymptomatic tears left alone.
- **Meniscal cyst** — lump below joint line, intermittent pain → arthroscopic decompression + partial meniscectomy.

Osteochondritis Dissecans (OCD)

- Avascular bone+cartilage fragment separates → loose body. Cause: **trauma**; site **80% medial aspect of lateral femoral condyle** (classically described as medial femoral condyle in many texts — lecture says lateral). Male **15–20 yrs**.
- **Wilson's sign** — pain on extending the medially-rotated flexed knee, relieved by lateral rotation.
- Treat: **stable/intact cartilage** → **rest 6–12 mo** (small heal spontaneously); unstable small → arthroscopic removal + drilling; large (>1 cm) → fix with pins/screws; defects → osteochondral plugs/chondrocyte sheets.

Loose Bodies

- Causes: injury chip, OCD, OA (cartilage/osteophyte), **Charcot**, synovial chondromatosis.
- **Sudden locking without injury** (varying position); "**joint mouse**" slips on palpation.
- Treat: remove arthroscopically if symptomatic (unless severe OA).

Deformities

- Normal knee = **5–7° valgus**. Bow legs (intercondylar distance) / knock knees (intermalleolar distance).
- **Physiological** — bow legs in babies, knock knees in 4-yr-olds; reassure, review 6-monthly. Operate only if marked by age 10 (>6 cm or >8 cm).
- **Pathological** — rickets, skeletal dysplasia, physeal injury.
- **Genu recurvatum** — congenital posture, joint laxity, RA, polio, trauma.

Osteoarthritis of the Knee

- **Commonest large joint affected**. Predisposing: injury, torn meniscus, instability, deformity.
- Often bilateral with **Heberden's nodes** (generalised OA); worst in **medial compartment**.
- Features: **age >50, overweight, bow-leg**, pain worse on use/stairs, stiffness after rest, swelling, crepitus.
- **X-ray standing AP** — joint space narrowing (one compartment), subchondral sclerosis/cysts, osteophytes, chondrocalcinosis.
- Treat: conservative (stick, **quads exercises**, analgesia, steroid injection, glucosamine/hyaluronan); operative — arthroscopic washout, **realignment osteotomy**, **replacement arthroplasty** (older, progressive).

Knee Swellings

- **Acute haemarthrosis** (immediate after injury) → suspect **ACL tear** if no fracture; also haemophilia.
- **Septic arthritis** — hot, tense; **Staph** (gonococcal in adults); aspirate pus, ↑ESR/WBC; urgent antibiotics ± surgical drainage if no response in 36 h.
- Chronic — OA, RA, synovial chondromatosis, PVNS, TB.

Walk-Away Points

- Ligaments stabilise the knee; **MRI for ligaments, standing AP X-ray for OA**.
- **Knee dislocation = vascular emergency** (40% popliteal injury).
- Extensor rupture site is **age-specific**; meniscus heals only in the **outer third**; **locking = bucket-handle tear**.
- **OCD** → **Wilson's sign**; loose body → "joint mouse" + locking.
- Knee = **commonest large-joint OA**; haemarthrosis after injury = ACL until proven otherwise.

Common Knee Problems — Revision Table	
Joint basics	Detail
Type	Hinge synovial; ligaments = main stabiliser
Symptoms	Pain, catching, clicking, locking
Imaging	MRI ligaments , standing AP X-ray for OA
Patellar dislocation	Detail
Direction	Lateral (valgus alignment)
Treatment	Conservative 2–3 wk; surgery if MPFL torn
Knee dislocation	Detail
Ligaments	ACL+PCL+LCL+MCL ruptured
Vascular	40% popliteal artery, 20% peroneal nerve
Treatment	Immediate reduction (no hyperextension), ex-fix if vascular
Extensor rupture by age	Site
Adolescent	Tibial tubercle avulsion
Young adult	Patellar tendon rupture
Middle-aged	Patella fracture
Older/ill	Quadriceps tendon rupture
Meniscus	Detail
Medial vs lateral	Medial torn more (less mobile)
Healing	Only outer third (vascular)
Sign	Locking = bucket-handle tear
Treatment	Arthroscopic meniscectomy/repair
OCD	Detail
Who	Male 15–20; trauma
Sign	Wilson's sign
Treatment	Stable→rest 6–12mo; unstable→fix/remove
Deformity	Detail
Normal	5–7° valgus
Physiological	Bow legs (baby), knock knees (4yr) — reassure
Operate	If marked by age 10
Knee OA	Detail
Note	Commonest large-joint OA
Who	>50, overweight, bow-leg
X-ray	Narrowing, sclerosis, cysts, osteophytes
Treatment	Stick, quads, steroid; osteotomy/replacement
Swelling	Cause
Acute haemarthrosis	ACL tear , haemophilia
Septic	Staph (adult: gonococcal); urgent drainage
Chronic	OA, RA, PVNS, TB

Bold = high-yield.